

Distributed Systems Programming

A.Y. 2025/26

Special Projects Proposals

1) Investigation of solutions for the automatic generation of server stubs from the REST API specification (1 or more students)

As shown in the first laboratory activity of the Distributed Systems Programming course, the development of a REST API server can be partially automated. After all the REST APIs are designed and specified in a document, their specification is used to automatically generate a server stub, which can serve as a skeleton for the API implementation. In fact, this server stub already provides a working interface for REST clients, so that the developers of the API implementation should only focus on the actual logic and functionality of each API described in the specification.

The tool that has been used for the server stub generation in the laboratory activity is Swagger Editor, which is compatible with the OpenAPI specification language. However, as seen during the lab activity, the server stubs generated by Swagger Editor have several limitations.

This project involves investigating alternative solutions to Swagger Editor for the automatic generation of server stubs from REST API OpenAPI specifications. The main tasks of the project are as follows:

- a) The students will search for possible solutions (one per student), and will try them to generate server stubs and implementations, starting from the specification of the REST APIs designed in the first laboratory activity.
- b) Complete documentation should be produced, describing the main characteristics of the identified solutions, explaining how they can be used to generate server stubs, and drawing considerations about their applicability in the next editions of the Distributed Systems Programming course in light of the results achieved within the practical experiments of the previous project task.

2) Design of REST APIs for MITO (max 1 student)

MITO (Multi-attack Impact-aware Transient Optimizer for Firewall Reconfiguration) is an automatic security service that computes the ordering of firewall reconfiguration updates that mitigate cyberattacks by prioritizing those with the highest impact. The framework implementing this service is written in Java, but it lacks REST APIs for the interaction with a human user or other services.

This project consists of improving and optimizing the capabilities of MITO by completing the following tasks:

- c) design JSON schemas for the representation of the core data structures required by the MITO service, as now only XML is supported;
- d) design the REST APIs for the MITO service and create their OpenAPI specification;
- e) implement the REST APIs for the MITO service, by connecting the APIs to the already existing implementation.

A documentation of all the choices taken for the design of the REST APIs, and some automated tests for the implementation of the REST APIs must also be provided as part of the solution.